Multimedia Signal Processing - 2nd Module

February, 23 2016

1. **Question (6 pts)**

   Consider two discrete memoryless sources X and Y. You observed the output of X and Y obtaining the following sequences:

$$X_1 = \{-1.7,\ -1.6,\ -1.9,\ \ 0.2,\ 0.5,\ 0.3,\ \ 2.2,\ 2.5,\ 2.1,\ -1.9,\ -1.6,\ -1.8,\ 2.2,\ 2.4,\ 2.4,\ 0.5,\ 0.5,\ 0.2,\ \ 2.3,\ \ 0.5\},$$
$$Y_1 = \{-1.5,\ \ 0.4,\ \ 2.4,\ -1.7,\ 0.0,\ 2.3,\ -1.8,\ 0.5,\ 2.3,\ -1.5,\ -1.5,\ -1.9,\ 0.3,\ 0.1,\ 0.4,\ 0.1,\ 0.3,\ 0.1,\ -1.6,\ -1.6\}.$$

   You then observed again the output of X and Y, obtaining two additional output sequences:

$$X_2 = \{0.5,\ 1.2,\ \ \ 3\},$$
$$Y_2 = \{1.1,\ 0.5,\ -1.4\}.$$

   (a) Compute the quantized sequences $\tilde{X}_1 = Q(X_1)$ and $\tilde{Y}_1 = Q(Y_1)$ using the following scalar uniform quantizer:

$$a_1 = -\infty,\ a_2 = -1,\ a_3 = 1,\ a_4 = +\infty,$$

$$Q(x) = I_k,\ \text{if } x \in (a_k, a_{k+1}]. \qquad I_k = \begin{cases} a_{k+1} - \frac{\Delta}{2}, & \text{if } k = 1, \\ \frac{a_k + a_{k+1}}{2}, & \text{if } k = 2, \\ a_k + \frac{\Delta}{2}, & \text{if } k = 3. \end{cases}$$

   (b) From $\tilde{X}_1$ and $\tilde{Y}_1$, estimate the joint p.m.f. $P(\tilde{X}; \tilde{Y})$ of a source that emits pairs of symbols $\{\tilde{X}(n), \tilde{Y}(n)\}$.

   (c) Based on the estimated p.m.f., compute the entropies $H(\tilde{X})$, $H(\tilde{Y})$ and $H(\tilde{X}; \tilde{Y})$. From these quantities, derive the mutual information $I(\tilde{X}; \tilde{Y})$.

   (d) Based on the estimated p.m.f., design the Huffman code that jointly encodes one symbol of $\tilde{X}$ and one symbol of $\tilde{Y}$. Compute the average Huffman code length and verify that it satisfies the theoretical bound.

   (e) Using the same quantizer described in (a), compute the quantized sequences $\tilde{X}_2 = Q(X_2)$ and $\tilde{Y}_2 = Q(Y_2)$. Consider again a source that emits pairs of symbols $\{\tilde{X}(n), \tilde{Y}(n)\}$. How many bits are used to encode the information contained in $\tilde{X}_2$ and $\tilde{Y}_2$ using Huffman coding scheme.

2. **Question (6 pts)**

   (a) Explain which are the common visual irrelevancy and redundancy effects that can be exploited for image coding.

   (b) Illustrate the block diagram of the JPEG compression coding scheme, detailing each block.

   (c) Which are the advantages of using DCT with respect to DFT? And with respect to KLT?

3. **Question (6 pts)**
   (a) Given a scalar uniform quantizer, derive the optimum reconstruction levels to minimize the error variance.
   (b) Illustrate the steps of the Lloyd-Max algorithm for scalar quantization, commenting the difference with respect to the optimum uniform quantizer described in (a).

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider two discrete memoryless sources (DMS) X and Y
   - [T] [F] $H(X) < 1$ if X is binary.
   - [T] [F] $H(X) + H(Y) = H(X;Y)$ if X and Y are independent.
   - [T] [F] $H(X) = 2$ means that each symbol of X can be represented with 2 bits.
   - [T] [F] $I(X;Y)$ cannot be computed for independent sources.

   (b) Consider a transform coding scheme $\mathbf{y} = \mathbf{T}\mathbf{x}$, where $\mathbf{T}$ is the matrix representing a unitary transform, $\mathbf{x}$ is the data vector, and $\mathbf{y}$ contains the transform coefficients
   - [T] [F] Transform coding gain is always positive.
   - [T] [F] To increase coding gain it is better to use a transform that grants uniformly distributed variance among the transform coefficients.
   - [T] [F] The optimal amount of bits for some transform coefficient may be theoretically negative.

   (c) Consider an audio coding system that exploits perceptual coding
   - [T] [F] A tone at time t_2 can mask signal components at time $t_1 < t_2$.
   - [T] [F] Critical bandwidths are linearly distributed in the Bark scale.
   - [T] [F] Perceptual coding is used for lossless audio compression.

5. **Question (Matlab - 10pts)**
   (a) Load the file `data.mat`, which contains the monodimensional signal `y` and the dictionary `dict` composed by 8 words (i.e., 8 rows) of 4 samples each.
   (b) Encode `y` using PCM with a uniform scalar quantizer with quantization intervals $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.
   (c) Encode `y` using a closed-loop DPCM scheme that works on blocks of 4 samples. The predictor of each block is the previous reconstructed block. The first block is transmitted using PCM. Use quantization intervals $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.
   (d) Encode `y` using a custom predictive coding scheme that works on blocks of 4 samples. Each block is predicted using the closest word in the dictionary in the MSE sense. The decoder knows the dictionary. For each block, the encoder sends to the decoder: (i) the index $w \in \{1, ..., 8\}$ of the selected word; (ii) the quantized prediction residuals. Quantize prediction residuals using $\Delta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ as quantization intervals.
   (e) For each case (i.e., (b), (c), (d)) plot the rate-distortion (RD) curves, using MSE as noise metric, SNR in decibels as distortion, and assuming that you are able to represent quantized coefficients using the minimum amount of bits. For point (c) assume that you are able to transmit the word indexes w with a lossless scheme that uses the minimum amount of bits.
   (f) Comment the obtained RD curves (hint: keep in mind the DPCM coding gain equation).